

MAR657 Visual Analytics

Individual Assignment 3

Chintan S. Jikkar

[U01989702]

```
library(dplyr)

library(ggplot2)

data <- read.csv("/Users/chintanjikkar/Desktop/PACE/Fall 25/Visual Analytics/
Data/marketing_campaign.csv")
head(data)
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	
##	1	5524	1957	Graduation	Single	58138	0	0	9/4
##	2	2174	1954	Graduation	Single	46344	1	1	3/8
##	3	4141	1965	Graduation	Together	71613	0	0	8/21
##	4	6182	1984	Graduation	Together	26646	1	0	2/10
##	5	5324	1981	PhD	Married	58293	1	0	1/19
##	6	7446	1967	Master	Together	62513	0	1	9/9

	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts
##	1	58	635	88	546	172
##	2	38	11	1	6	2
##	3	26	426	49	127	111
##	4	26	11	4	20	10
##	5	94	173	43	118	46
##	6	16	520	42	98	0

	MntGoldProds	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases	
##	1	88	3	8	10
##	2	6	2	1	1
##	3	42	1	8	2
##	4	5	2	2	0

```
## 5          15          5          5          3
## 6          14          2          6          4
##   NumStorePurchases NumWebVisitsMonth AcceptedCmp3 AcceptedCmp4 AcceptedCm
p5
## 1          4          7          0          0
0
## 2          2          5          0          0
0
## 3         10          4          0          0
0
## 4          4          6          0          0
0
## 5          6          5          0          0
0
## 6         10          6          0          0
0
##   AcceptedCmp1 AcceptedCmp2 Complain Response
## 1          0          0          0          1
## 2          0          0          0          0
## 3          0          0          0          0
## 4          0          0          0          0
## 5          0          0          0          0
## 6          0          0          0          0

data$Income <- as.numeric(gsub("[$,]", "", as.character(data$Income)))
```

Univariate Non-Graphical Analyses

Categorical: Education

```
table(data$Education)
```

Education	Count
Basic	54
Graduation	1,116
Master	365
2n Cycle	200
PhD	481

Most customers have a Graduation degree, showing a highly educated base.

Quantitative: Income

```
summary(data$Income)
```

Statistic	Value
Min.	1,730
1st Qu.	35,303
Median	51,382
Mean	52,247
3rd Qu.	68,522
Max.	666,666

Average income is around \$52,000, with most incomes below \$100,000 and one extreme outlier.

Multivariate Non-Graphical Analyses

Categorical vs Categorical: Education vs Marital Status

```
table(data$Education, data$Marital_Status)
```

Education	Absurd	Alone	Divorced	Married	Single	Together	Widow	YOLO
2n Cycle	0	0	23	80	36	56	5	0
Basic	0	0	1	20	18	14	1	0
Graduation	1	1	119	429	246	285	35	0
Master	1	1	37	138	75	102	11	0
PhD	0	1	52	190	96	116	24	2

Graduation is the leading education level among all marital statuses, especially Married and Together.

Quantitative vs Quantitative: Income vs MntWines

```
cor(data$Income, data$MntWines, use="complete.obs")
```

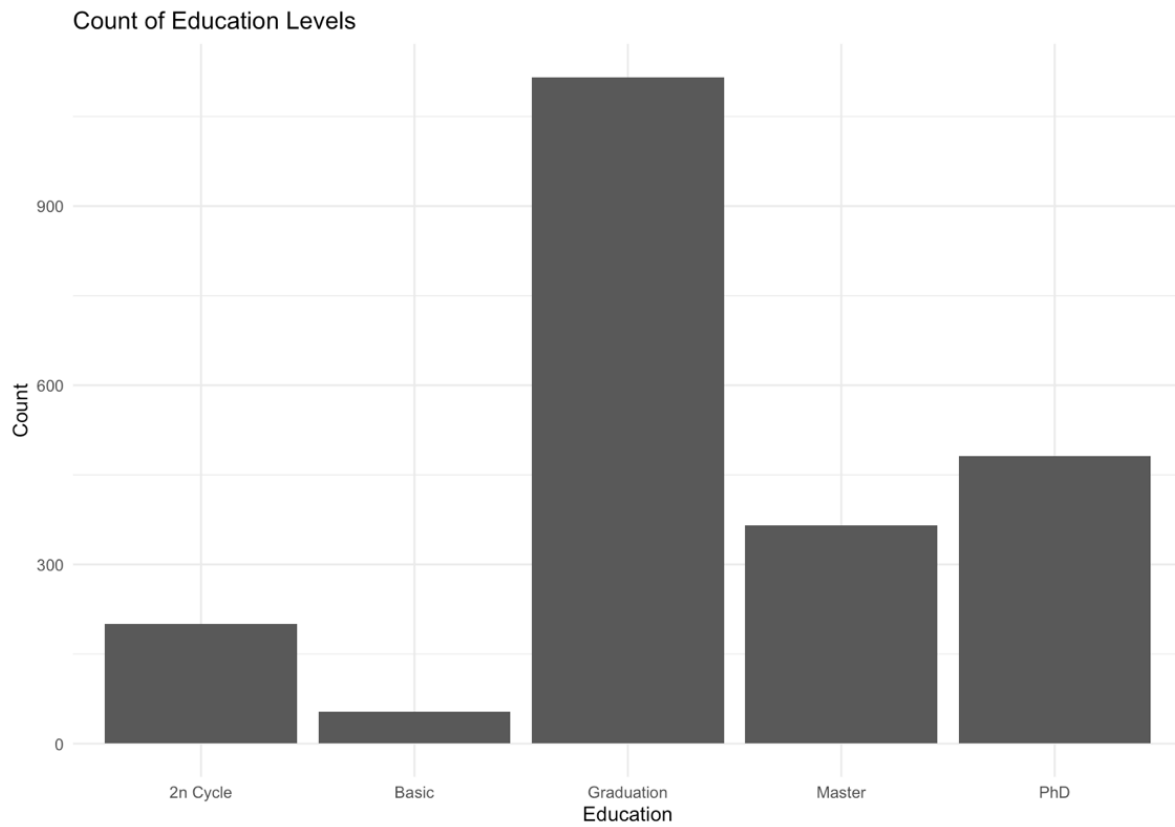
$$r = 0.58$$

Higher income is moderately correlated with greater wine spending.

Univariate Graphical Analyses

Categorical: Bar Plot of Education

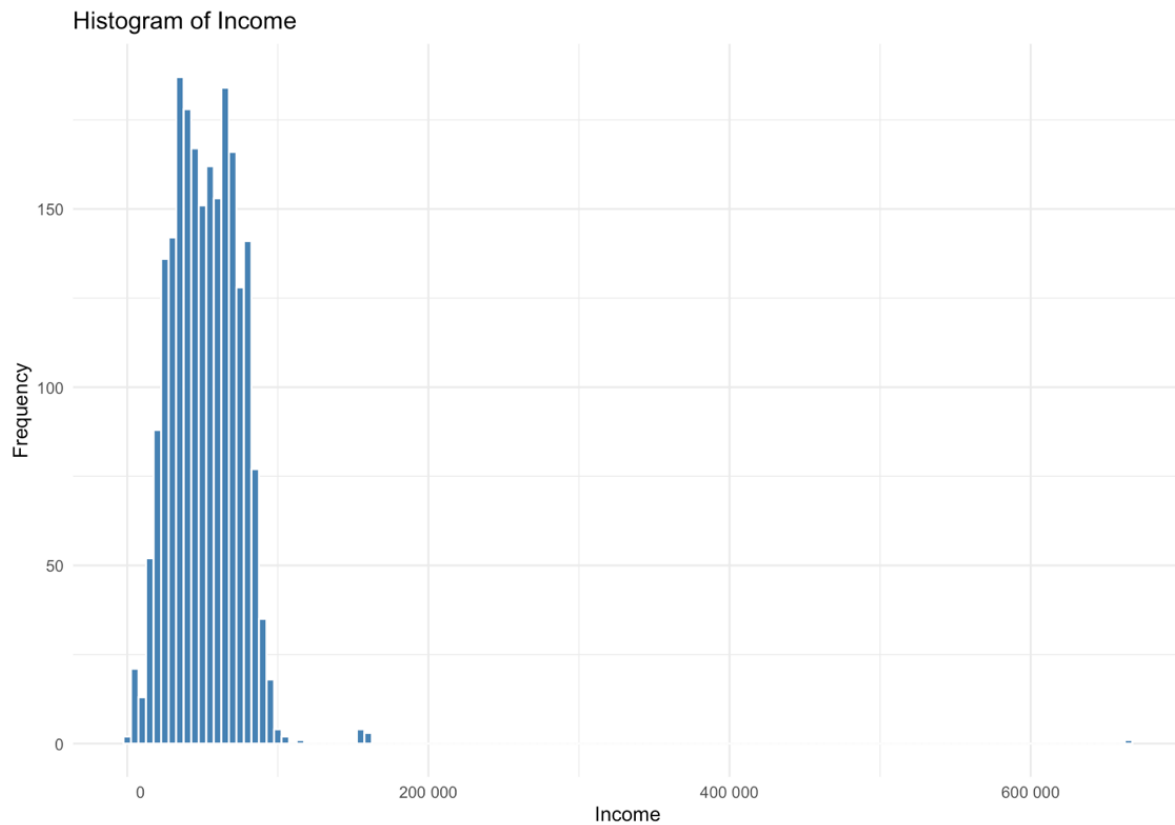
```
ggplot(data, aes(x = Education)) +  
  geom_bar() +  
  theme_minimal() +  
  labs(title = "Count of Education Levels", x = "Education", y = "Count")
```



Graduation dominates the education categories in the bar chart.

Quantitative: Histogram of Income

```
ggplot(data, aes(x = Income)) +  
  geom_histogram(binwidth = 5000, fill = "steelblue", color = "white") +  
  theme_minimal() +  
  labs(title = "Histogram of Income", x = "Income", y = "Frequency") +  
  scale_x_continuous(labels = function(x) format(x, big.mark = " ",  
scientific = FALSE))
```

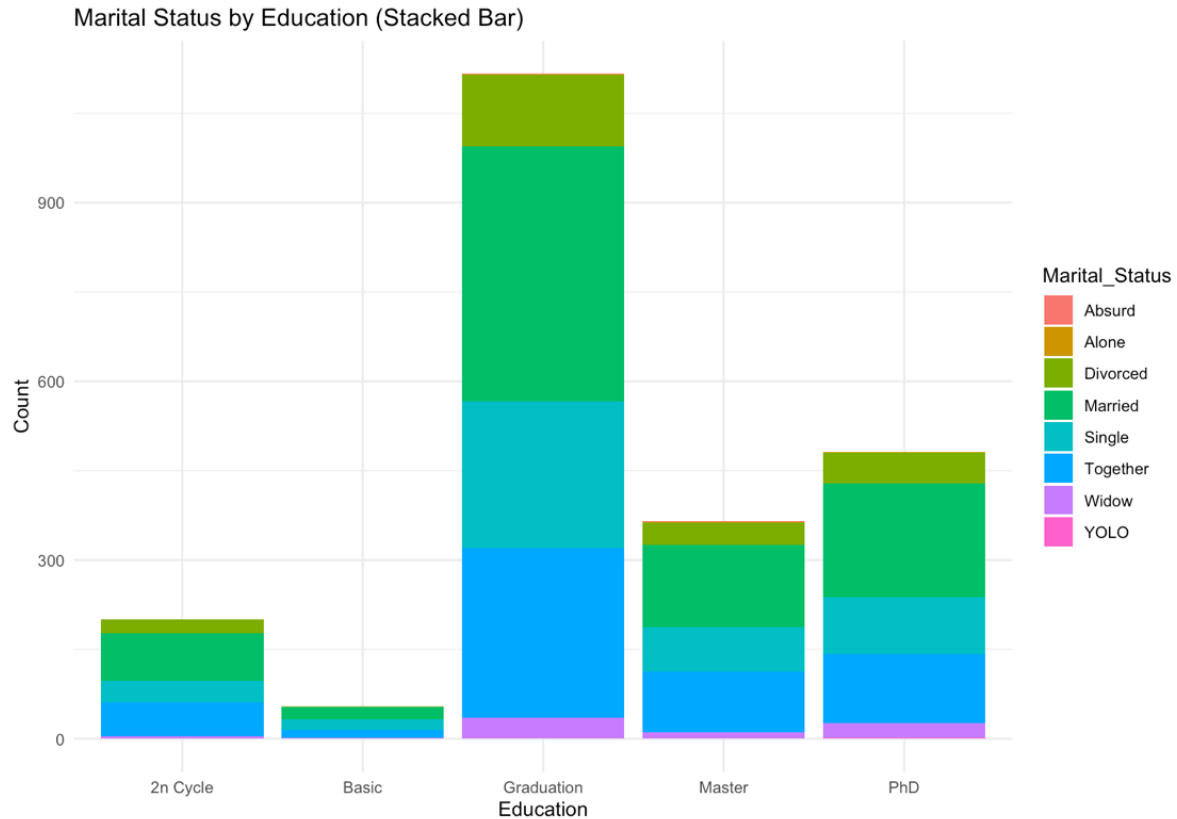


Income distribution is skewed right, with few customers at very high income levels.

Multivariate Graphical Analyses

Categorical vs Categorical: Stacked Bar of Marital Status by Education

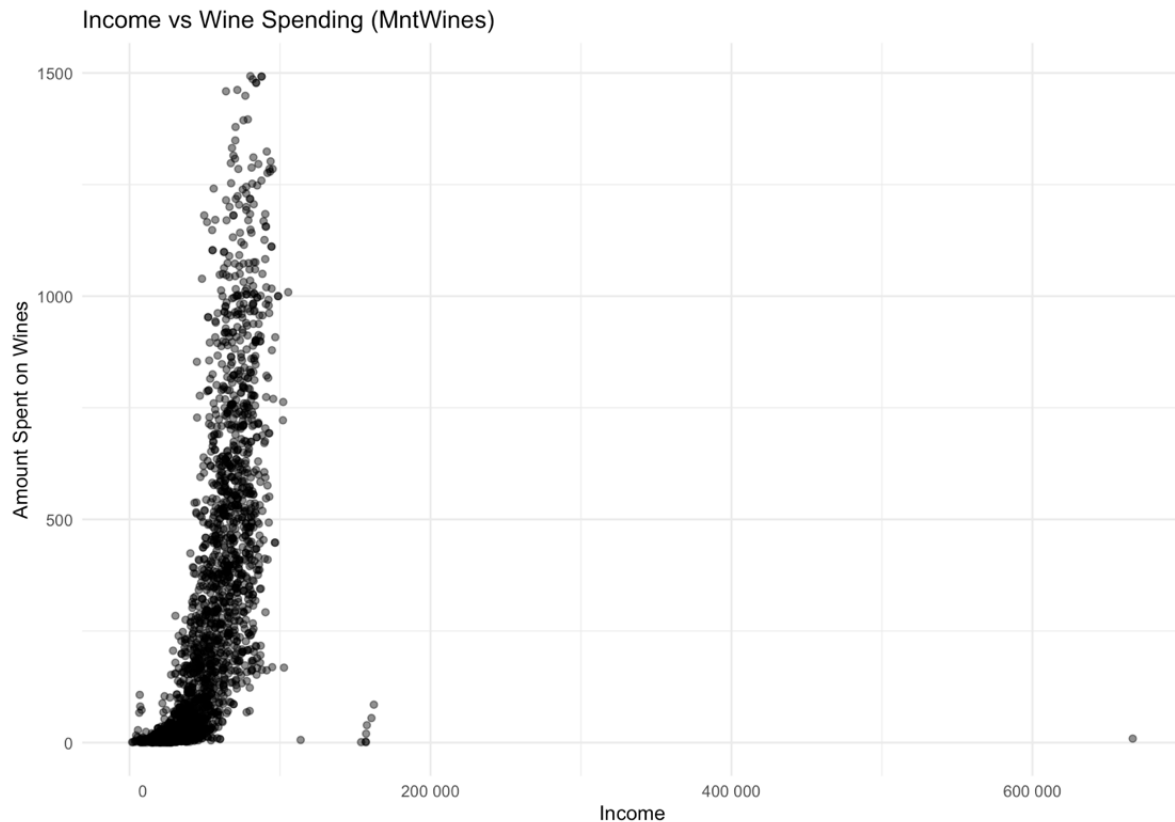
```
ggplot(data, aes(x = Education, fill = Marital_Status)) +  
  geom_bar(position = "stack") +  
  theme_minimal() +  
  labs(title = "Marital Status by Education (Stacked Bar)", x = "Education",  
y = "Count")
```



Higher education groups are mostly Married or Together in the stacked bar plot.

Quantitative vs Quantitative: Scatterplot of Income vs Wine Spending

```
ggplot(data, aes(x = Income, y = MntWines)) +  
  geom_point(alpha = 0.5) +  
  theme_minimal() +  
  labs(title = "Income vs Wine Spending (MntWines)", x = "Income", y =  
"Amount Spent on Wines") +  
  scale_x_continuous(labels = function(x) format(x, big.mark = " ",  
scientific = FALSE))
```



Scatterplot reveals most wine spending occurs among mid-to-high income customers, supporting the correlation.